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Spurious sea ice formation caused by oscillatory ocean tracer advection schemes

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ABSTRACT

Tracer advection schemes used by ocean models are susceptible to artificial oscillations: a form of numerical error whereby the advected field alternates between overshooting and undershooting the exact solution, producing false extrema. Here we show that these oscillations have undesirable interactions with a coupled sea ice model. When oscillations cause the near-surface ocean temperature to fall below the freezing point, sea ice forms for no reason other than numerical error. This spurious sea ice formation has significant and wide-ranging impacts on Southern Ocean simulations, including the disappearance of coastal polynyas, stratification of the water column, erosion of Winter Water, and upwelling of warm Circumpolar Deep Water. This significantly limits the model's suitability for coupled ocean-ice and climate studies. Using the terrain-following-coordinate ocean model ROMS (Regional Ocean Modelling System) coupled to the sea ice model CICE (Community Ice CodE) on a circumpolar Antarctic domain, we compare the performance of three different tracer advection schemes, as well as two levels of parameterised diffusion and the addition of flux limiters to prevent numerical oscillations. The upwind third-order advection scheme performs better than the centered fourth-order and Akima fourth-order advection schemes, with far fewer incidents of spurious sea ice formation. The latter two schemes are less problematic with higher parameterised diffusion, although some supercooling artifacts persist. Spurious supercooling was eliminated by adding flux limiters to the upwind third-order scheme. We present this comparison as evidence of the problematic nature of oscillatory advection schemes in sea ice formation regions, and urge other ocean/sea-ice modellers to exercise caution when using such schemes.

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1. Introduction

A central element of ocean models is the advection of temperature and salinity, simulated using a number of different numerical methods (Griffies et al., 2000). These advection schemes are susceptible to various types of numerical error (Hecht et al., 2000; Shchepetkin and McWilliams, 1998; Lilly, 1965), including issues with stability, artifical dissipation (by which water masses overmix), and artifical oscillations. It is the last such issue that we focus on here. Oscillations, also known as overshoots or dispersion, are characterised by tracer fields that appear jagged and erratic after advection, with false extrema (Shchepetkin and McWilliams, 1998). These oscillations most likely occur near steep gradients in the given tracer field, which can be poorly resolved at low resolution.

Oscillatory behaviour in various tracer advection schemes has been well-studied (Hecht et al., 2000; Shchepetkin and

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McWilliams, 1998; Pietrzak, 1998), and its potential for undesirable feedbacks with ocean processes has been demonstrated (Hecht, 2010; Farrow and Stevens, 1995; Gerdes et al., 1991). However, there are no published investigations of how oscillatory behaviour interacts with coupled ocean/sea-ice models. When simulating regions of sea ice formation, there is a major threshold associated with the freezing point. Oscillations which cause the ocean temperature to fall below the freezing point therefore have physical significance beyond simple numerical error, as acknowledged by Hecht et al. (2000). In some ways this situation is similar to the simulation of regions with strong freshwater inflow, where oscillations could cause negative salinity.

In this study, we use a terrain-following-coordinate ocean model (Shchepetkin and McWilliams, 2005; Galton-Fenzi et al., 2012) with a coupled sea ice model (Hunke et al., 2015) to show that oscillatory tracer advection schemes have a significant impact on sea ice formation. When oscillations cause the ocean temperature to fall below the freezing point, this spurious supercooling is then removed from the near-surface layers as frazil ice. This frazil forms even if the ocean is already shielded from atmospheric heat fluxes by a layer of solid sea ice. As a result, unphysically thick patches of sea ice occur. These thick patches of sea ice have a significant influence on other physical processes, including coastal polynyas, stratification of the water column, dense water formation, and the properties of deep water masses. Note that despite the similar terminology, this phenomenon is distinct from the temporal oscillations in ice-ocean Ekman transport discussed by Roberts et al. (2015).

Steep horizontal gradients in tracer fields are more common in the local coordinate space of sigma- or terrain-following-coordinate ocean models (Griffies et al., 2000), which discretise the vertical dimension using fractional depth of the water column rather than absolute depth. Here we use “horizontal” to describe a line or surface of constant vertical level on the terrain-following grid, rather than of constant depth. If the underlying bathymetry is steep, horizontally adjacent grid cells can lie at quite different depths. A sharp gradient in depth often translates to a sharp gradient in tracers such as temperature or salinity. By contrast, the same region modelled with z-coordinates would have weaker temperature and salinity gradients between horizontally adjacent grid cells, which would by definition lie at exactly the same depth. Since the ocean is, for the most part, well mixed in the horizontal but stratified in the vertical, there are more opportunities for sharp horizontal gradients in sigma-space than in z-space. As a consequence, sigma-coordinate and terrain-following coordinate ocean models may be particularly susceptible to artificial oscillations over areas of steep bathymetry.

Advection schemes are typically tested on idealised domains, often at very high resolution and sometimes with reduced dimensionality. In practice, however, advection schemes are ultimately incorporated into “realistic configurations”, which we define here as three-dimensional forward models on observed domains, generally with rougher bathymetry and lower resolution than idealised setups. We believe it is valuable to communicate the effects of numerical error on realistic configurations, since they are the ones most often used to understand observations, make future projections, and ultimately inform policy. By understanding how errors present themselves in realistic domains, with realistic forcing and commonly used resolution, we hope to forewarn other members of the ocean modelling community who might otherwise experience similar problems, and to provide an acceptable solution.

2. Advection schemes

In a finite-volume discretisation of the primitive equations for ocean circulation, the concentration of a tracer at a given grid box represents the volume average over that grid box. The advection of the tracer depends on the advective fluxes through each face of the grid box. However, the calculation of these fluxes depends on the area-averaged concentration of the tracer over each face, and there is no exact solution for the interpolation of these values from the adjacent volume-averages. Several different numerical methods have been developed to address this interpolation, giving rise to different advection schemes. In our simulations we compare three different advection schemes from the standard distribution of ROMS (the Regional Ocean Modelling System (Shchepetkin and McWilliams, 2005)), as well as a limiter scheme designed to remove oscillations from one of the advection schemes.

2.1. Centered fourth-order

The centered fourth-order advection scheme (Shchepetkin and McWilliams, 2005) interpolates tracers to grid box faces using a midpoint-average modified by a gradient or curvative term. This interpolation is centered, and fourth-order accurate, in space. While this scheme is less prone to oscillations than its second-order counterpart (Shchepetkin and McWilliams, 1998), oscillations still occur (Leonard and Mokhtari, 1990).

2.2. Akima fourth-order

The Akima fourth-order advection scheme (Shchepetkin and McWilliams, 2005) differs from the centered fourth-order scheme only in its calculation of the gradient or curvature term, which utilises harmonic averaging rather than a simple midpoint average. Shchepetkin and McWilliams (2005) found that this scheme reduces oscillations compared to the centered fourth-order scheme.

2.3. Upwind third-order

There are several different schemes in the upwind third-order family; here we use the scheme known as UTOPIA (Shchepetkin and McWilliams, 2005; Rasch, 1994; Leonard, 1993). The interpolation of tracers to grid box faces is not centered in space, but rather is biased toward the upwind or upstream direction, which depends on the sign of the given velocity component. Upwind schemes in general suppress oscillations (Griffies et al., 2000), as the truncation errors associated with upwind interpolation are dominated by dissipation rather than dispersion. In first-order upwind schemes, this can lead to unphysical diapycnal mixing which breaks down fronts between water masses. In comparison, third-order schemes exhibit significantly reduced artificial dissipation (Leonard, 1993). The residual is considered “implicit diffusion” which can often be compensated for by reducing or even eliminating explicitly parameterised diffusion (Farrow and Stevens, 1995; Dinniman et al., 2015). Note that we use this scheme only for horizontal tracer advection; it is paired with the centered fourth-order advection scheme in the vertical.

2.4. Upwind limiters

While the upwind third-order scheme is known to significantly reduce oscillations compared to centered advection schemes, oscillations can still occur near sharp gradients (Leonard, 1993). These oscillations can be targeted and removed using a flux limiter scheme. Here we implement the “universal limiter” developed by Leonard and Mokhtari (1990) as part of the ULTRA-SHARP scheme; see also Norris (2000). This scheme calculates limits for the interpolated tracer values at each grid box face, and by enforcing these limits it clips false extrema, as follows:

Let C^* be the concentration of a tracer interpolated by the upwind third-order advection scheme to a given grid box face in the

east-west direction. Let C_{WW}, C_W, C_E, and C_{EE} denote the volume-averaged (i.e. known) concentrations two cells west, one cell west, one cell east, and two cells east of this face. The flux limiters are then used to calculate a new concentration, C, as follows:

$$C = \begin{cases} \text{median}[C_E, C^*, \text{median}[C_W, C_E, C_{WW} + \alpha(C_W - C_{WW})]] & \text{if } u \geq 0 \\ \text{median}[C_W, C^*, \text{median}[C_E, C_W, C_{EE} + \alpha(C_E - C_{EE})]] & \text{if } u < 0 \end{cases} \quad (1)$$

where u is the zonal velocity at the grid box face. The equations are similar for the north-south faces. As with the baseline upwind third-order advection scheme, the upwind limiters are only applied in the horizontal.

Eq. (1) defines three bounds on C, the first two of which are fixed, and the third which is modified by the parameter α which controls the amount of clipping. Smaller values of α lead to stricter clipping of oscillations, but may also clip some legitimate extrema. The specific choice of α is determined empirically. The original reference (Leonard and Mokhtari, 1990) set $\alpha = 10$, while we use $\alpha = 5$. Our solution showed little sensitivity to values of α greater than 5, and the smaller value is seen as the more conservative design choice.

Further details on the upwind limiter scheme and its implementation in ROMS can be found in an upcoming paper by Brassington et al. (2017).

2.5. Other advection schemes

Another popular tracer advection scheme among ROMS users is MPDATA (Multidimensional Positive Definite Advection Transport Algorithm (Smolarkiewicz, 1984)). This scheme is sign-preserving, and therefore unsuitable for temperature gradients across 0 °C. In order to use MPDATA for the simulation of sea ice formation regions or ice shelf cavities, the temperature field would need to be transformed to ensure positivity. The advection could be done in Kelvins rather than degrees Celsius; alternatively, a smaller value surpassing the minimum freezing point (such as 5 °C, noting the depth-dependent freezing point in ice shelf cavities) could be added to temperature before advection and subtracted after.

We experimented with the HSIMT scheme (High-order Spatial Interpolation at the Middle Temporal level (Wu and Zhu, 2010)), a non-oscillatory scheme that shares many properties with MPDATA, but is suitable for negative temperatures. Unfortunately, this scheme caused unacceptably high levels of spurious diapycnal mixing in our configuration (not shown). For different configurations, perhaps with higher resolution, HSIMT may be suitable. For all advection schemes, errors of both dissipative and dispersive nature are less likely at high resolution.

3. Model description

Our simulations are performed with the MetROMS model, which consists of the ROMS ocean model (Regional Ocean Modelling System (Shchepetkin and McWilliams, 2005)) including ice shelf thermodynamics (Galton-Fenzi et al., 2012), coupled to the CICE sea ice model (Community Ice CodE (Hunke et al., 2015)) using the MCT coupler (Model Coupling Toolkit (Larson et al., 2005; Jacob et al., 2005)). The coupling infrastructure was implemented by the Norwegian Meteorological Institute (Debernard et al., 2017), and is described in Section 3.5.

3.1. Domain

We run MetROMS on a circumpolar Antarctic domain, with a northern boundary at 30 °S. Horizontal resolution for both ROMS and CICE is $\frac{1}{4}^\circ$ scaled by cosine of latitude, and the South Pole is relocated to ensure approximately equal resolution around the coastline. This results in Cartesian resolutions ranging from 8 to 10 km on the Antarctic continental shelf to 15–20 km in the Antarctic Circumpolar Current (ACC). While this resolution is higher than most global coupled models, it is lower than many ROMS simulations, which often use smaller domains (Galton-Fenzi et al., 2012; Cougnon et al., 2013; Gwyther et al., 2014). Bathymetry, ice shelf draft, and land/sea masks were given by RTopo 1.05 (Timmermann et al., 2010), and smoothed as in Lemarié et al. (2012) to ensure stability with terrain-following coordinates.

3.2. Forcing

ROMS and CICE are forced with the same atmospheric state, from which each component calculates bulk fluxes of heat, salt, and momentum. Over grid cells containing a mix of sea ice and open water, these fluxes are merged by the coupler.

Atmospheric forcing is provided by the ERA-Interim reanalysis (Dee et al., 2011) and consists of 6-hourly fields for near-surface temperature, pressure, humidity, winds, and total cloud cover; and 12-hourly fields for rain, snow, and evaporation. The 6-hourly fields are linearly interpolated to each model timestep. The 12-hourly fields represent total water fluxes over the given 12-h period, and are not interpolated in time but rather applied at a constant rate with a step change every 12 h. For our one-year simulations we use 1992 forcing. We also apply an estimate of freshwater fluxes from iceberg melt, using the monthly-averaged 100-year climatology simulated by Martin and Adcroft (2010).

ROMS also requires lateral boundary conditions at the open northern boundary, 30 °S. For this we use monthly averaged fields of temperature, salinity, and horizontal velocity from the ECCO2 cube92 reanalysis (Menemenlis et al., 2008; Wunsch et al., 2009) for 1992. Northern boundary conditions for sea surface height are given by the AVISO climatology (AVISO, 2011). The numerical treatment of these boundary conditions is discussed in Section 3.3.

3.3. Ocean configuration

ROMS is a split-explicit, free-surface, terrain-following-coordinate ocean model (Shchepetkin and McWilliams, 2005). We use the development version 3.7 of the Rutgers ROMS code, with the addition of ice shelf thermodynamics using the standard three-equation parameterisation (Galton-Fenzi, 2009; Galton-Fenzi et al., 2012). Our grid has 31 terrain-following vertical levels, with higher resolution near the surface and bottom of the water column. Typical vertical resolutions are 1–3 m at the surface and 200–300 m in the deep interior ocean. In ice shelf cavities, vertical resolution is often finer than 1 m. With such thin vertical layers we require a baroclinic timestep of 5 min, with 30 barotropic timesteps for each baroclinic.

While we vary the tracer advection scheme in our experiments, advection of momentum uses the same scheme for all experiments: upwind third-order advection in the horizontal, and centered fourth-order advection in the vertical. Subgrid-scale mixing is parameterised using biharmonic viscosity (for momentum) and Laplacian diffusivity (for tracers). The horizontal mixing coefficients are 1000 m^4/s for biharmonic viscosity, and either 15 or 150 m^2/s for Laplacian diffusivity (varying by experiment, see Section 3.7). These coefficients apply to the resolution of the largest grid cell (approx. 24 km) and are scaled linearly for smaller grid cells. We use the Large-McWilliams-Doney interior closure scheme (Large et al., 1994) which includes the KPP boundary layer parameterisation.

Particular care must be taken at the open northern boundary, where instabilities can easily occur due to discontinuities be-

tween the lateral boundary conditions and the simulated fields. We follow the method used by the Southern Ocean State Estimate simulations (Mazloff et al., 2010) on a similar circumpolar domain (Matthew Mazloff, personal communication). First, zonal velocity u is clamped to zero at the northern boundary to prevent waveguide artifacts. Second, the bathymetry is modified to be constant in y over the northernmost 15 rows of grid cells (approx. 3° latitude). In this way, boundary conditions for meridional velocity v can smoothly transition into the domain without immediately encountering variations in topography. Finally, a sponge layer is applied over these northernmost 15 rows, whereby the diffusivity coefficient linearly increases to 10 times its background value at the northern boundary, and the viscosity coefficient (which has units of m^4/s rather than m^2/s) to 100 times the background value. The numerical methods used to apply the northern boundary conditions consist of the Chapman scheme for sea surface height (Chapman, 1985), the Flather scheme for barotropic v (Flather, 1976), and the radiation-nudging scheme for baroclinic v, temperature, and salinity (Marchesiello et al., 2001).

3.4. Sea ice configuration

CICE is a dynamic-thermodynamic sea ice model which has been successfully coupled to several global climate models (Hunke et al., 2015). We use version 5.1.2 of the CICE code. Our simulations have 7 vertical ice layers plus 1 snow layer, and 5 ice thickness categories. The sea ice timestep is 30 min for both dynamic and thermodynamic processes.

Our configuration uses the "mushy" thermodynamics scheme (Turner et al., 2013). We use the level-ice melt pond parameterisation with Stefan refreezing. Radiation is treated by the Delta-Eddington scheme (Briegleb and Light, 2007).

For dynamics, we use the elastic-viscous-plastic scheme (Hunke and Dukowicz, 1997) as revised by Bouillon et al. (2013). Sea ice transport follows an incremental remapping approach Lipscomb and Hunke (2004). We use the ice strength formulation of Rothrock (1975), with the ridging participation and redistribution functions of Lipscomb et al. (2007).

3.5. Coupling

ROMS and CICE run on separate processors, and communication between the two models is facilitated by the MCT coupler (Larson et al., 2005; Jacob et al., 2005). Since the same horizontal grid is used for the ocean and sea ice, the coupling is relatively straightforward. The only issue comes from the fact that ROMS uses the Arakawa C-grid while CICE uses the Arakawa B-grid (Arakawa and Lamb, 1977). That is, while ROMS and CICE share the same horizontal grid cells, the location of variables within that cell (i.e. at the centre, corners, or edges) is not the same. However, both the B- and C-grid have tracers in the centre of each cell. All coupling is therefore done on the tracer grid, with variables linearly interpolated to and from the centre of each cell where required.

ROMS and CICE exchange fields every 30 min of simulation (i.e., every sea ice timestep). All fields are averaged or integrated over the preceding 30-min coupling interval. CICE passes ROMS the sea ice concentration, heat and salt fluxes, the shortwave radiation coming through the ice, and the ice-ocean stress vector. ROMS passes CICE the sea surface temperature and salinity, ocean velocity averaged over the upper 5 m, sea surface height, and the freeze-melt potential.

The last variable warrants particular attention given the subject of this paper. Freeze-melt potential is the energy flux (in W/m^2) associated with the temperature difference from the freezing point, and is used by the sea ice model to form frazil ice (> 0) or melt existing sea ice (< 0). We follow the same algorithm as the POP (Parallel Ocean Program) z-coordinate ocean model Smith et al. (2010) to calculate freeze-melt potential, with minor modifications to account for the differing vertical coordinate system.

Freeze potential is integrated over the upper 5 m of the water column, removing supercooling in these layers every ocean timestep. If warmer layers (above freezing) overlay colder layers, they are allowed to melt the frazil ice formed below. The surface layer is also allowed to melt frazil ice from previous ocean timesteps, within the same coupling interval. Any additional energy flux in the surface layer available to melt ice is accumulated each timestep as melt potential (< 0).

The salinity fluxes associated with both frazil ice formation and basal melting are calculated by CICE and returned to the ocean model as a surface salt flux. This design is required to ensure salt conservation, since the mushy thermodynamics scheme in CICE allows sea ice salinity to evolve. Also to match the mushy thermodynamics in CICE, the freezing point T_f (in °C) is given by

$$T_f = \frac{S}{-18.48 + \frac{18.48}{1000} S} \tag{2}$$

where S is salinity in psu. Note that we neglect the depth-dependence of the freezing point, since this term is negligible over the upper 5 m. Freeze-melt potential is not calculated in ice shelf cavities, which are masked out of the CICE domain. In ice shelf cavities the depth-dependence of the freezing point is vital, and supercooling is handled differently. For a full explanation see Galton-Fenzi (2009), Galton-Fenzi et al. (2012).

3.6. Initialisation

ROMS is initialised with temperature and salinity from the ECCO2 cube92 reanalysis (Menemenlis et al., 2008; Wunsch et al., 2009) for January 1992. These fields are extrapolated into ice shelf cavities using a nearest-neighbour method in Cartesian space. Initial velocity and sea surface height are set to zero.

Sea ice is initialised using January 1992 observations from the NOAA/NSIDC Climate Data Record of Passive Microwave Sea Ice Concentration (Meier et al., 2013). Wherever observed sea ice concentration exceeds 15%, we intialise with concentration 100%, sea ice thickness of 1 m, and snow thickness of 0.2 m. This method of sea ice initialisation is similar to that used by Kjellsson et al. (2015).

3.7. Experiments

We found that short simulations were sufficient to illustrate the impact of advection errors on sea ice formation and deep water masses. Therefore, each simulation we present here ran for one year under 1992 forcing. We ran six different simulations, varying the advection scheme, level of parameterised diffusion, and presence of upwind limiters, as summarised and named in Table 1.

Since the upwind third-order advection scheme has implicit diffusion, unlike the fourth-order centered and Akima schemes, we ran the latter two schemes with two levels of parameterised diffusion: low (Laplacian diffusivity of 15 m^2/s, the same as the upwind schemes) and high (150 m^2/s). These values are scaled by grid size, as described in Section 3.3, meaning the diffusivity coefficient for a typical grid cell will be smaller.

In the following results, we treat U3_LIM as a baseline simulation to which all other simulations are compared. The presence of upwind limiters ensures there will be no spurious supercooling in this simulation. Note also that the upwind limiters have a negligible computational overhead, as shown in Table 1. Activating limiters in U3_LIM only causes a 3% increase in walltime compared to the U3 simulation.

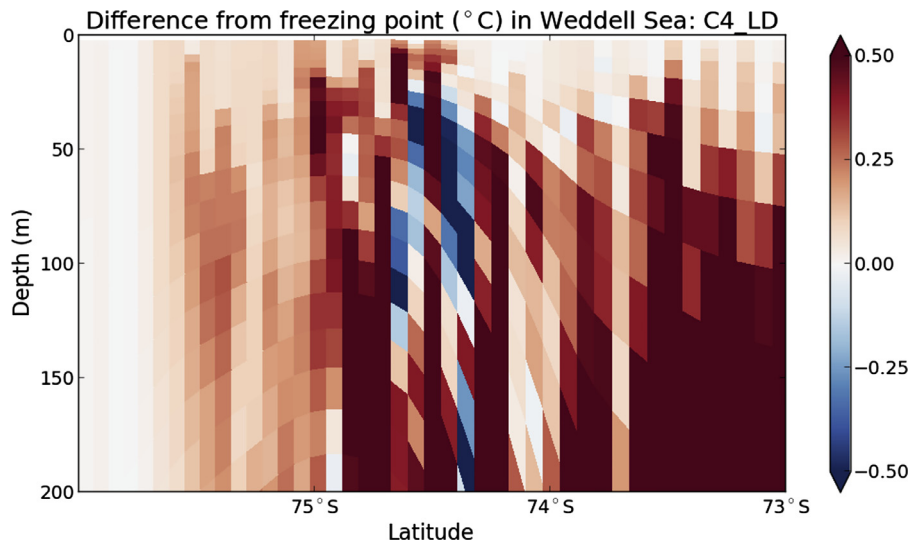

Fig. 1. Difference from the surface freezing point, i.e. $T - T_f$, where T is ocean temperature and T_f is the surface freezing point as given by Eq. (2). Values are shown on a single i-slice of the model grid through the Weddell Sea, for a single timestep on 26 June during the C4_LD simulation. In this region the grid is significantly rotated from regular longitude-latitude axes; longitudes in this plot range from 35 °W in the southernmost cells to 40 °W in the north.

Table 1
Summary of simulations performed. Upwind limiters can only be combined with the upwind third-order scheme. Walltime refers to the hours and minutes required to complete each 1-year simulation. Simulations were run on the Raijin cluster of the Australian National Computational Infrastructure, using 608 cores (512 ocean + 96 sea ice) which are Intel Xeon Sandy Bridge/Broadwell 2.6 GHz.

Experiment name	Tracer advection scheme	Laplacian diffusivity (m^2/s)	Upwind limiters	Walltime h:mm
U3_LIM	Upwind third-order	15	yes	4:32
U3	Upwind third-order	15	no	4:23
C4_LD	Centered fourth-order	15	n/a	4:22
A4_LD	Akima fourth-order	15	n/a	4:30
C4_HD	Centered fourth-order	150	n/a	4:30
A4_HD	Akima fourth-order	150	n/a	4:25

4. Results and discussion

4.1. Supercooling

In an accurate simulation of the Southern Ocean, we would not expect to see any supercooling below 5 m depth, except possibly a small amount transported from the sub-ice shelf cavities. Since the ocean is cooled only at the surface, and ROMS removes supercooling from the upper 5 m every 5 min, there should not be time for supercooling to propagate to deeper water masses. However, if these deeper water masses are near the freezing point, oscillatory advection errors could cause spurious supercooling.

An example of this phenomenon in experiment C4_LD is shown in Fig. 1, which plots the difference from the freezing point at each cell in a slice through the model grid, at a single timestep. With no time-averaging, spatial averaging, or spatial interpolation, the presence of oscillatory advection errors is obvious. Rather than a smooth field, the model output appears noisy and discontinuous. Isolated columns of subsurface supercooling - some exceeding 0.5 °C below the freezing point - are surrounded by much warmer water. Over a steep region of the continental slope, around 74.5 °S, this supercooling extends as deep as 200 m.

Note that advection errors causing spurious formation of sea ice do not show up in Fig. 1, since this supercooling is removed from the upper 5 m of the water column as soon as it forms. Nonetheless, this figure illustrates the existence of oscillations and their unphysical nature. The salinity field (not shown) exhibits similar discontinuities, indicating that oscillations can deterioriate simulated fronts across the continental slope. Even disregarding the impact on sea ice formation, this behaviour presents a barrier to the accurate simulation of the interior ocean.

4.2. Sea ice formation

Frazil formation of sea ice should only occur where there is some amount of open water (i.e. sea ice concentration less than 100%), since supercooling of the ocean is driven by atmospheric heat fluxes. Again, there may be a small contribution from supercooled water exiting ice shelf cavities, but this should be restricted to grid cells adjacent to ice shelf fronts. Congelation, whereby seawater freezes directly onto the sea ice base due to conduction of heat through the ice, is calculated separately from frazil ice formation and at no stage involves supercooling in ROMS. Therefore, in an accurate simulation we would expect to see simulated frazil formation restricted to regions of open water, such as coastal polynyas and the northernmost edge of sea ice extent during the freeze season.

The U3_LIM simulation performs reasonably well in this regard. Annually averaged frazil ice formation (Fig. 2(a)) is strongest in coastal polynyas, particularly in the Ross and Weddell Seas. Areas of weaker frazil formation offshore correspond to the northernmost sea ice extent at different times of the year. Other simulations display significantly higher frazil formation than U3_LIM, particularly on the continental shelf, and sometimes far from the coast. These anomalies are most apparent for the C4_LD and A4_LD simulations (Fig. 2(c) and (e)), and tend to line up with steep bathymetric features such as the continental shelf break.

Increasing the parameterised diffusion for the C4_HD and A4_HD simulations smoothes out many of the oscillations, and therefore removes most of the spurious frazil production (Fig. 2(d) and (f)). However, there are still significant anomalies on the continental shelf. The upwind third-order scheme U3 with no limiters (Fig. 2(b)) behaves similarly.

These anomalies in frazil ice formation for each simulation are reflected in the sea ice thickness. At the time of maximum sea ice area (23 August), the U3_LIM simulation displays a realistic distribution of effective sea ice thickness (Fig. 3(a)), with thicker ice from the continental shelf exported by the Ross and Wed-

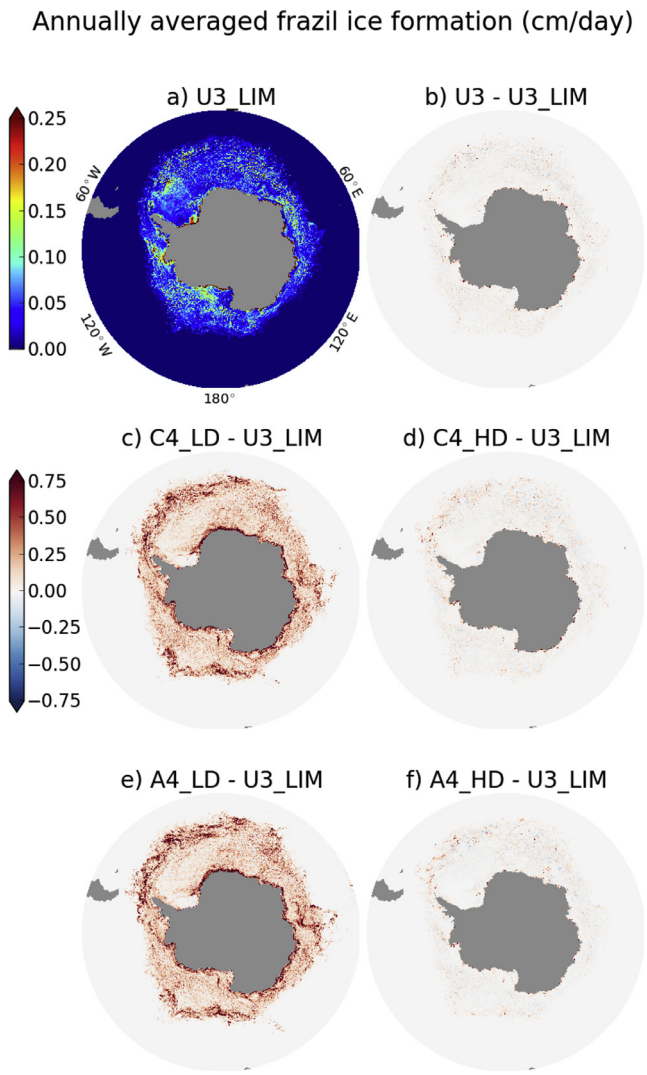

Fig. 2. Annually averaged frazil ice formation during each 1-year simulation. Absolute frazil formation (cm/day) for the U3_LIM simulation is shown in (a); anomalies from U3_LIM for other simulations are shown in (b)–(f). Experiment names are explained in Table 1.

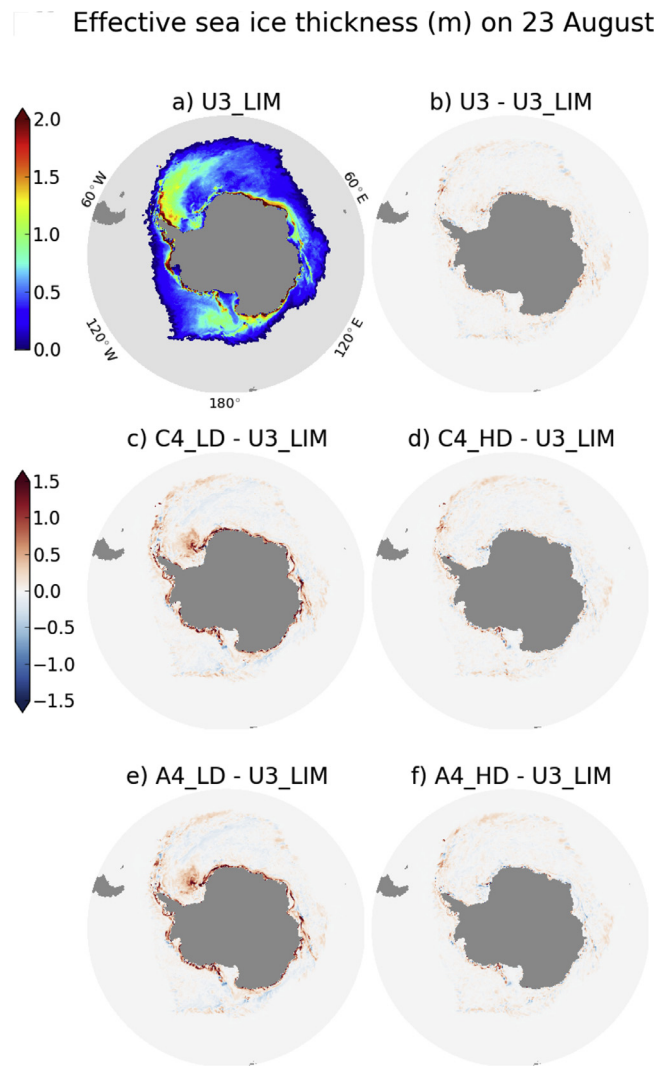

Fig. 3. As Fig. 2, but showing effective sea ice thickness (concentration multiplied by height at each grid cell) on 23 August (daily average), which corresponds to the day of maximum sea ice area in the U3_LIM simulation. Cells where sea ice concentration is below 15% have been masked out of (a).

dell Gyres, and thinner ice in polynya regions along the coastline. The C4_LD and A4_LD simulations (Fig. 3(c) and (e)), by contrast, show thick packs of sea ice along the continental shelf break. Sea ice thickness in the offshore Weddell Sea is approximately tripled compared to the U3_LIM simulation. The regions of increased sea ice thickness differ somewhat from the regions of anomalous frazil formation seen in Fig. 2, due to sea ice drift. However, given that other sea ice thermodynamic growth terms (congelation and snow-to-ice flooding) are similar in all six simulations, the thick ice is indeed driven by frazil formation.

These thick packs of sea ice are mostly eliminated by increasing the parameterised diffusion (C4_HD and A4_HD simulations; Fig. 3(d) and (f)). However, thicker ice still exists along the continental shelf break of the Amundsen and Bellingshausen Seas, extending into the eastern Ross Sea. In other regions, such as offshore of the Amery Ice Shelf, sea ice is actually thinner than in U3_LIM. The upwind third-order scheme with no limiters (U3; Fig. 3(b)) exhibits slightly thicker sea ice than U3_LIM along most of the continental shelf break.

4.3. Stratification

Sea ice which forms due to oscillatory advection errors alone will melt as soon as it drifts away from the spurious supercooling into warmer waters, which can occur in as little as 1 grid cell. In regions of persistent spurious supercooling causing thick sea ice, the rate of melting is significant. The resulting large freshwater flux stratifies the ocean, decreasing the mixed layer depth and suppressing deep convection. Note that sea ice melt cools, as well as freshens, the ocean surface.

Sea surface salinity at the sea ice maximum (23 August) is shown for the U3_LIM simulation in Fig. 4(a). The other simulations (Fig. 4(b) through 4(f)) exhibit significant fresh anomalies (> 1 psu) in regions of spurious sea ice formation. For the C4_LD and A4_LD simulations (Fig. 4(c) and (e)), these regions are so large that the meltwater freshens almost the entire domain. Freshening is particularly significant on the continental shelf, as well as east of the Drake Passage where meltwater from the Weddell Sea is swept into the ACC. For the C4_HD, A4_HD, and U3 simulations (Figs. 4(b), (d), and (f)), this meltwater is restricted to the northern ACC east of the Drake Passage, as well as small patches on the western side of the Antarctic Peninsula.

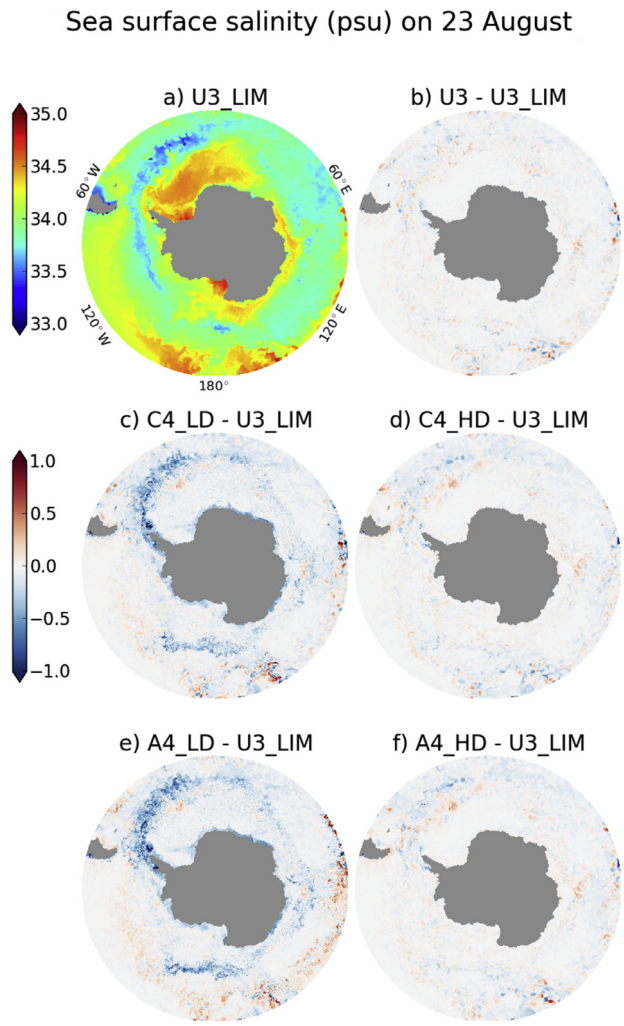

Fig. 4. As Fig. 3, but for sea surface salinity. Note that ice shelf cavities are masked.

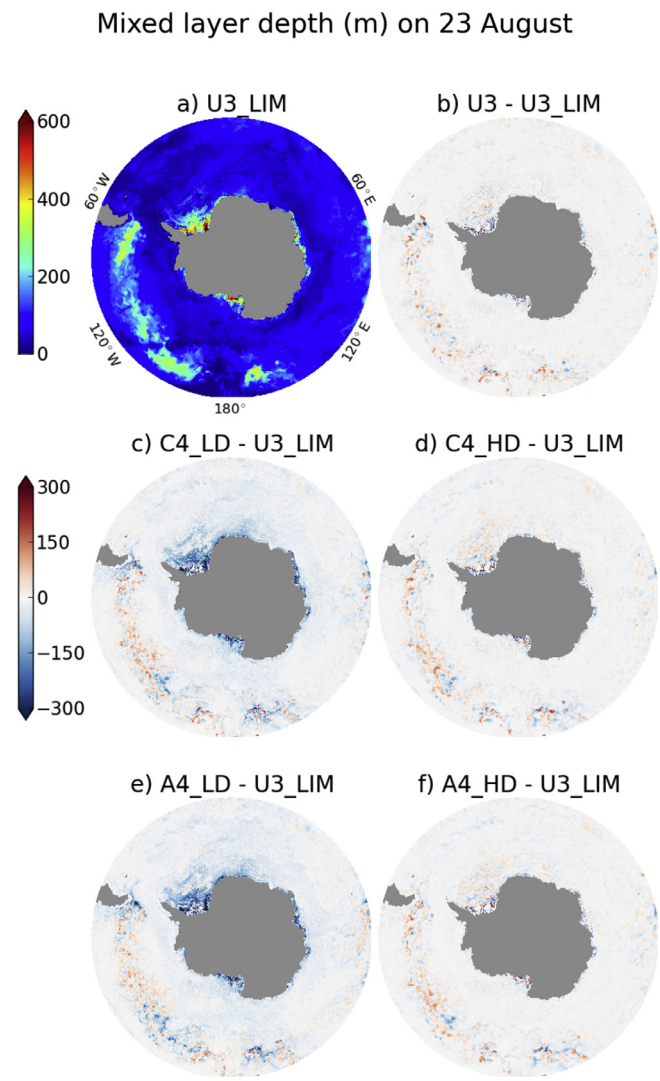

Fig. 5. As Fig. 3, but for mixed layer depth (depth of the oceanic boundary layer as calculated by the KPP parameterisation). Note that ice shelf cavities are masked.

Mixed layer depth, defined here as the depth of the oceanic boundary layer as calculated by the KPP parameterisation (Large et al., 1994), is significantly affected by this surface freshening. During the sea ice maximum on 23 August, the U3_LIM simulation has deep mixed layers (> 300 m) in known regions of dense water formation (Ohshima et al., 2013), most prominently the Ross and Weddell Seas as well as Prydz Bay near the Amery Ice Shelf (Fig. 5(a)). The C4_LD and A4_LD simulations (Fig. 5(c) and (e)) have much shallower mixed layers in these regions, as well as throughout the Weddell Sea and most of the ACC.

This behaviour is less pronounced for the C4_HD and A4_HD simulations (Fig. 5(d) and (f)), although shallower mixed layers are still apparent in some regions of the continental shelf. No consistent anomalies are seen in mixed layer depth for the U3 simulation (Fig. 5(b)).

4.4. Coastal polynyas

Simulations with significant spurious supercooling are characterised by an almost complete absence of year-round coastal polynyas. Polynyas are regions of open water created by the strong southerly katabatic winds, which push sea ice away from the coast. As new sea ice forms in its place, the resulting brine rejection triggers deep convection, and warmer water from below upwells to the surface.

Spurious formation of sea ice suppresses coastal polynyas in two ways:

1. Dynamic: Thick sea ice forms due to spurious supercooling above steep bathymetry, particularly the continental shelf break. These thick packs of sea ice decrease the mobility of upstream sea ice, which often encompasses known coastal polynya regions. Export of sea ice from polynya regions is therefore inhibited.
2. Thermodynamic: The stabilising effect of meltwater from the transport of these thick patches of sea ice, as discussed in the previous section, suppresses deep convection in polynya regions. Upwelling of warmer water from below, which is necessary to prevent polynyas from immediately freezing over, is thereby suppressed.

Year-round coastal polynyas are present during the U3_LIM simulation in many regions along the Antarctic coastline. Fig. 6(a) shows sea ice concentration for this simulation near the Amery Ice Shelf front (67°E–86°E) during the sea ice maximum on 23 August. Substantial polynyas are present in Barrier Bay (≈ 80°E), with several smaller polynyas dotted along the coastline.

These polynyas are significantly reduced in size during the C4_LD simulation (Fig. 6(b)), with sea ice concentration near 1 al-

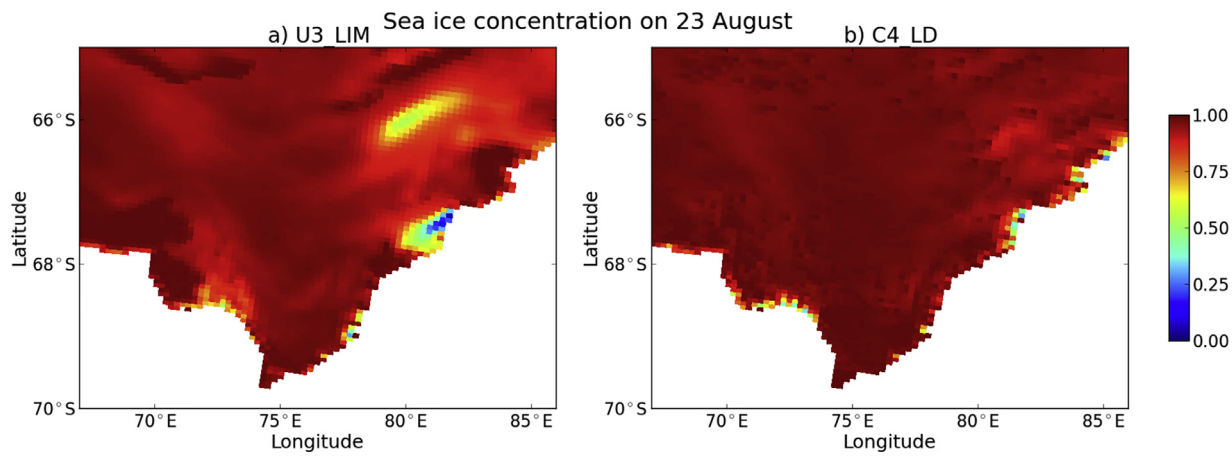

Fig. 6. Sea ice concentration (fraction) on 23 August (daily average) near the Amery Ice Shelf front, including Barrier Bay (≈ 80°E). 23 August corresponds to the maximum sea ice extent in the U3_LIM simulation. Concentration for the U3_LIM simulation is shown in (a), and for the C4_LD simulation in (b). Experiment names are explained in Table 1.

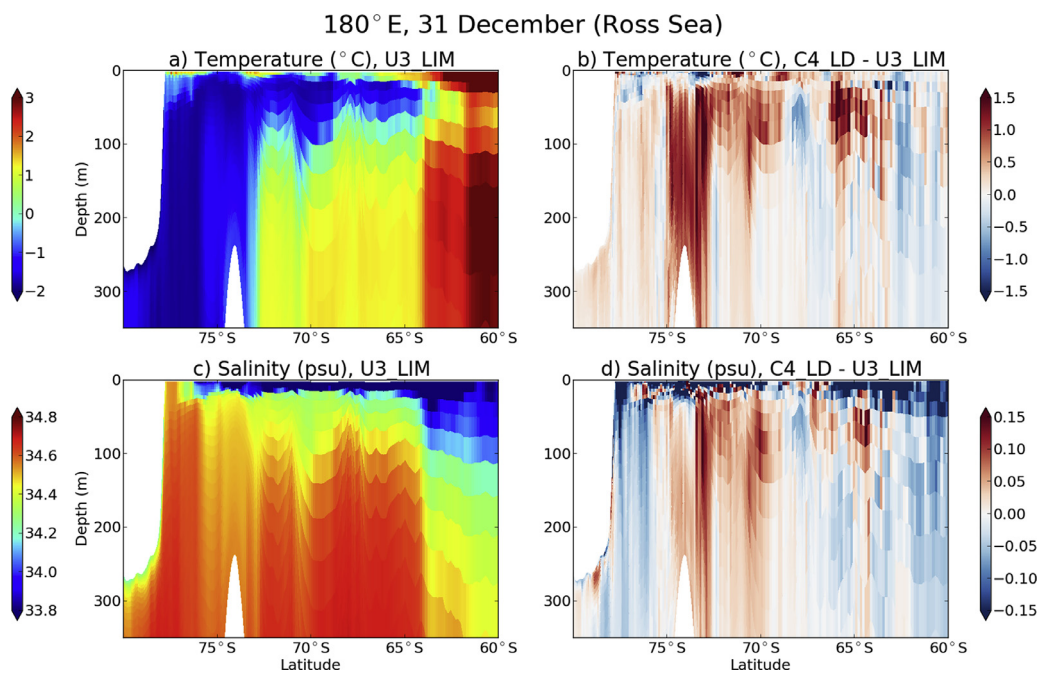

Fig. 7. Latitude vs. depth slices of temperature (a,b) and salinity (c,d), interpolated to 180°E, which intersects the Ross Sea. Ice shelf cavities are explicitly simulated by ROMS, and the Ross Ice Shelf front can be seen on the left side of each plot. Values are averaged over the last day of simulation, 31 December. The U3_LIM simulation (a,c) and anomalies for the C4_LD simulation with respect to U3_LIM (b,d) are shown. Experiment names are explained in Table 1.

most everywhere. The high levels of spurious sea ice formation seen in C4_LD, and the resulting buttressing and stratifying effects, have nearly eliminated the coastal polynyas.

4.5. Deep water masses

As well as shutting down coastal polynyas, the stratification discussed in Section 4.3 causes the subsurface Winter Water layer to erode. Winter Water is the remnant of the winter mixed layer, overlaying warmer Circumpolar Deep Water and forming a subsurface temperature minimum. If the winter mixed layer is shallow, as seen in the C4_LD and A4_LD simulations, the thickness of this year-round layer is reduced.

Significant impacts are seen after only one year. Fig. 7 shows zonal slices of temperature and salinity interpolated to 180 °E (through the Ross Sea) on the last day of the U3_LIM simulation (31 December), and anomalies for the C4_LD simulation. A layer of Winter Water (temperature ≈ −0.5 °C, salinity ≈ 34.4 psu) is visible in the U3_LIM simulation (Figs. 7(a) and (c)), with a tongue of cold water extending into the subsurface ACC, and warm salty Circumpolar Deep Water (temperature > 0 °C, salinity > 34.5 psu (Jacobs, 2004)) below 100 m depth in the offshore water column. Dense water formation is also active in the Ross Sea polynya, visible as a column of uniform salinity (≈ 34.6 psu) immediately north of the Ross Ice Shelf front.

The Winter Water layer has been partially replaced by Circumpolar Deep Water in the C4_LD simulation (Fig. 7(b) and (d)), which exhibits significant warm and saline anomalies above 100 m in the offshore water column. Warming of approximately 1 °C is apparent over the continental slope (near 74°S), as Modified Circumpolar Deep Water begins to travel onto the continental shelf. Additionally, fresh anomalies north of the ice shelf front, concentrated near the surface, indicate a shutdown of dense water formation in this region due to stratification.

In longer simulations with C4_LD (not shown), Modified Circumpolar Deep Water eventually makes its way into ice shelf cavities, including the Ross. Area-averaged ice shelf melt rates dramatically increase, well beyond observational estimates, as warm water floods into the cavity. In contrast to these indirect effects of spurious supercooling on ice shelf melt rates, we do not see any significant evidence of direct effects, i.e. spurious supercooling causing refreezing onto the ice shelf base. Within ice shelf cavities, the ocean is shielded from the motion of wind and sea ice at the surface, and ocean velocities are therefore much slower. Oscillatory advection errors are much less likely when velocities are near-zero.

4.6. Effect of coupling interval

In our experiments, ROMS and CICE exchange fields every sea ice timestep, equal to 30 min. Some global climate models, such as the Community Earth System Model (which includes CICE), instead have a sea-ice/ocean coupling interval of 24 h (Roberts et al., 2015). This daily averaging of coupling fields is computationally inexpensive, but it also filters semi-diurnal feedbacks such as oscillations in ice-ocean Ekman transport, which can have a significant impact on sea ice dynamics as demonstrated by Roberts et al. (2015). Long coupling intervals also introduce lags into the model which can lead to instabilities and chaotic behaviour.

We found that spurious sea ice formation was sensitive to the length of the ice-ocean coupling interval. Simulations with 24-h coupling intervals (not shown) experienced more significant anomalies than the 30-min simulations analysed here. For all advection schemes, frazil formation as well as sea ice basal melt was higher with 24-h coupling than with 30-min coupling, since the longer lags in the coupled system allowed both freeze potential and melt potential to overshoot. This compounded the effects of spurious supercooling in the C4_LD and A4_LD simulations. Anomalies in sea ice thickness, surface salinity, stratification, and dense water mass properties were more pronounced with 24-h coupling than with 30-min coupling. This indicates that long ice-ocean coupling intervals, as used by some global models, exacerbate spurious sea ice formation caused by oscillatory advection schemes.

5. Conclusions

We have shown that artificial oscillations in tracer advection schemes have undesirable interactions with a coupled sea ice model. Numerical oscillations cause spurious supercooling and sea ice formation, particularly over steep areas of bathymetry such as the continental shelf break, where thick packs of sea ice buildup. The dynamic and thermodynamic impacts of this thick sea ice suppress coastal polynyas, stratify the water column, and cause Winter Water to erode and Circumpolar Deep Water to shoal. The long-term significance of these biases for the simulated climate state is unknown and is not the subject of this study.

The centered fourth-order and Akima fourth-order horizontal tracer advection schemes were the worst performing, with significantly more incidents of spurious supercooling and sea ice formation than the upwind third-order scheme. Increasing the parameterised diffusion improved the performance of both fourth-order schemes, but supercooling artifacts were still present. Spurious supercooling was only eliminated with the inclusion of a flux limiter for the upwind third-order scheme. However, only minor differences were apparent between the upwind third-order scheme with and without limiters, indicating that the limiters did not need to be activated very frequently.

Upwind limiters are a particularly attractive solution for preventing oscillations in high-resolution ocean models, because they allow parameterised diffusion to be very low or even nonexistent. For eddy-permitting or eddy-resolving simulations, explicit mixing can be destructive to the simulation of these eddies. The high levels of parameterised diffusion that we find necessary to add to the centered fourth-order and Akima fourth-order schemes to suppress oscillations would therefore be undesirable. On the other hand, oscillations should be less likely at higher resolution due to better resolution of steep gradients in tracer fields. In this case, substantial increases in parameterised diffusion may not actually be necessary.

When artificial oscillations breach thresholds such as the freezing point, they can no longer be considered superficial numerical error. In the extreme case, as we have shown, they can have detrimental effects on model physics. These results may be model-dependent, and in particular may be influenced by the terrain-following coordinates in our configuration. Nonetheless, we suggest that other users of coupled ocean/sea-ice models exercise caution when using oscillatory advection schemes and watch carefully for signs of spurious sea ice formation.

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